



Mohammad Ruhul Amin (Rafi)

EEE Undergraduate · Drone Systems & Circuit Simulation

Profile

Dedicated 2nd-year Electrical and Electronic Engineering student at Jamalpur Science and Technology University (JSTU) specializing in Circuit Simulation, Control Systems, and Hardware Prototypes. Passionate about drone aerial technology, regenerative energy recovery, and engineering automated solutions that bridge digital logic with the physical world.

Projects & Research Experience

- 2024 — Present | Department of EEE, JSTU
Automated Water Tank Level Control System
Circuit Simulation Researcher (Matlab / Simulink)
 - Modeled closed-loop fluid reservoir level control using MATLAB & Simulink under Lecturer Md. Mahfuzur Hayan (Course EEE2108).
 - Simulated dynamic inflow pump actuation and real-time overflow protection cutoff.
 - Presented comprehensive simulation analysis as member of a 6-student team at JSTU.
- 2024 | EEE Laboratory, JSTU
Smart Regenerative Braking System Prototype
Hardware Prototype & Power Electronics Lead
 - Engineered physical hardware prototype capturing vehicle kinetic braking energy using mechanically coupled dual DC motor dynamo setup.
 - Integrated Arduino microcontroller, 5V relay switching, and DC-DC buck-boost voltage regulation to charge battery storage during deceleration.
 - Demonstrated live working prototype in EEE lab; produced verified video demonstration.
- 2023 — 2024 | Embedded Hardware Lab
Automatic Plant Watering & Smart Irrigation System
Embedded Systems & IoT Developer
 - Developed autonomous precision irrigation device with soil moisture sensor, mini submersible DC water pump, and 18650 Li-ion battery source.
 - Programmed automated moisture threshold switching via 5V relay module to conserve water and eliminate overwatering risks.
- 2023 — Present | Aerial Tech & Visuals
Drone Aerial Systems & Perspective Photography
Drone Enthusiast & Visual Storyteller
 - Investigating multirotor aerodynamics, flight controllers, and ESC brushless motors.
 - Curated 23 original visual perspective frames capturing landscapes across diverse rural and natural terrains of Bangladesh.

References

Department of Electrical and Electronic Engineering (EEE)

Jamalpur Science and Technology University (JSTU), Jamalpur, Bangladesh
Academic & Coursework References available upon request: info@jstu.ac.bd

CONTACT

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EDUCATION

B.Sc. in Electrical & Electronic Eng.

Jamalpur Sci. & Tech. Univ. (JSTU)
2nd Year (Session 2023 – 2027)

Higher Secondary Certificate (HSC)

Shahid Syed Nazrul Islam College
GPA: 5.00 / 5.00 (Golden A+)

Secondary School Certificate (SSC)

Mamun Smrity Public School
GPA: 5.00 / 5.00 (Golden A+)

SKILLS

- MATLAB & Simulink
- Circuit Simulation & Analysis
- Control Systems Modeling
- Drone Flight Dynamics
- Arduino & Embedded C/C++
- Coupled DC Dynamos & Relays
- Power Recovery Electronics
- Sensors & Transducers
- Aerial Cinematography

AWARDS & HONORS

- Academic Excellence
SSC & HSC Golden A+ 5.00/5.00
- EEE2108 Simulation Sessional
Top Presentation Team
- Robotics & Drone Systems
JSTU Technical Workshop